

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Mock Interview

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO5: *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.*
(BL5)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 25

Code of Conduct

I M. Meghana (192525435) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *M. Meghana*

Assessment Tasks

Mock Interview

Answer the following interview-oriented questions clearly and concisely. Support your responses with architecture-level reasoning wherever appropriate.

1. Explain the difference between HDFS and traditional file systems.
2. Why is replication important in distributed storage systems?
3. How does MapReduce divide work between map and reduce phases?
4. What role does YARN play in large-scale Hadoop processing?
5. Differentiate NAS and SAN in an enterprise data environment.
6. What is the purpose of ETL in a big data pipeline?
7. How does Apache NiFi help in data integration workflows?
8. What are the key failure points in a Hadoop cluster and how would you handle them?
9. How would you design a secure data ingestion pipeline for a large organization?
10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Mock Interview Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Knowledge	25%	Shows deep and accurate technical understanding	Good technical understanding	Basic understanding	Weak understanding
Problem Solving	20%	Responds with strong architecture-level reasoning	Reasonable reasoning	Basic reasoning	Weak reasoning
Communication	20%	Answers are confident, precise, and professional	Mostly clear communication	Moderate clarity	Weak communication
Practical Awareness	20%	Connects concepts to real-world implementation well	Some practical relevance	Limited practical relevance	No practical linkage
Confidence and Professionalism	15%	Highly confident and professional	Generally professional	Moderately professional	Low confidence and professionalism

Mock Interview – Hadoop & Big Data

1. Explain the difference between HDFS and traditional file systems.

Answer:

HDFS is a distributed file system designed to store very large datasets across multiple machines. Unlike traditional file systems that usually store data on a single server, HDFS divides files into blocks and distributes them across Data Nodes. It also provides replication for fault tolerance.

Architecture point: HDFS follows a Master–Worker architecture, where the Name Node manages metadata and Data Nodes store actual data.

2. Why is replication important in distributed storage systems?

Answer:

Replication creates multiple copies of data on different nodes. If one node fails, another replica can serve the data, preventing data loss and improving availability.

For example, with a replication factor of 3, HDFS maintains three copies of each data block on different Data Nodes.

Architecture point: Replication provides fault tolerance, high availability, and improved read performance.

3. How does MapReduce divide work between map and reduce phases?

Answer:

MapReduce divides processing into two major stages:

- **Map phase:** Reads input data and converts it into intermediate key-value pairs.
- **Shuffle and Sort:** Groups intermediate data according to keys.
- **Reduce phase:** Processes each key and its associated values to produce the final result.

For example, in word counting, the mapper generates (word, 1) and the reducer adds all values for the same word.

4. What role does YARN play in large-scale Hadoop processing?

Answer:

YARN stands for Yet Another Resource Negotiator. It manages cluster resources and schedules applications across Hadoop nodes.

Its main components are:

- **ResourceManager** – manages overall cluster resources.
- **NodeManager** – manages resources on individual nodes.
- **ApplicationMaster** – manages a particular application.

Architecture point: YARN separates resource management from data processing, allowing Hadoop to support multiple processing frameworks.

5. Differentiate NAS and SAN in an enterprise data environment.

Feature	NAS	SAN
Access	File-level	Block-level
Protocol	NFS, SMB	Fibre Channel, iSCSI
Usage	File sharing, documents	Databases, virtualization
Management	Relatively simple	More complex
Performance	Good	Generally higher

Interview answer:

NAS provides shared files over a network, while SAN provides block-level storage that servers can use like local disks. NAS is commonly used for file sharing, whereas SAN is preferred for high-performance enterprise workloads such as databases and virtualization.

6. What is the purpose of ETL in a big data pipeline?

Answer:

ETL means Extract, Transform, Load.

- **Extract:** Collect data from sources such as databases, applications, and logs.
- **Transform:** Clean, validate, filter, and convert the data.
- **Load:** Store the processed data in a target system such as a data warehouse or Hadoop-based storage.

Architecture point: ETL converts raw heterogeneous data into a consistent and analytics-ready format.

7. How does Apache NiFi help in data integration workflows?

Answer:

Apache NiFi is a data-flow automation and integration tool. It can collect data from multiple sources, route it, transform it, and deliver it to different destinations.

For example:

Database → NiFi → Data validation → Transformation → HDFS → Analytics

NiFi provides a visual flow-based interface, monitoring, back-pressure, provenance tracking, and integration with many data sources.

8. What are the key failure points in a Hadoop cluster and how would you handle them?

Answer:

Major failure points include:

1. **DataNode failure** – handled using HDFS replication and automatic re-replication.
2. **NameNode failure** – handled using **NameNode High Availability** with active and standby NameNodes.
3. **Network failure** – use redundant network paths and proper monitoring.
4. **Disk failure** – use multiple disks and replication.
5. **ResourceManager failure** – use YARN High Availability.
6. **Application/job failure** – configure retries and monitor failed tasks.

Architecture point: Hadoop follows a failure-aware design, where redundancy and automatic recovery minimize service disruption.

9. How would you design a secure data ingestion pipeline for a large organization?

Answer:

I would design the pipeline with security at every layer:

Data Sources → Secure Ingestion → Validation → Encryption → Storage → Processing → Monitoring

Key measures include:

- Use TLS/SSL for data in transit.
- Encrypt sensitive data at rest.
- Use authentication such as Kerberos.
- Apply role-based access control.
- Validate and sanitize incoming data.
- Maintain audit logs and data provenance.
- Use network segmentation and firewalls.
- Protect credentials using a secure secrets-management system.

Architecture point: Security should follow a defense-in-depth approach, rather than relying on a single security mechanism.

10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Answer:

I would explain it like this:

“Imagine we have a very large warehouse of information. Instead of keeping everything in one huge room, Hadoop divides the information across many smaller rooms. Multiple workers can then process different parts at the same time. If one worker fails, another copy of the information is available. When the amount of data increases, we can add more workers to the system.”

So, Hadoop provides distributed storage, parallel processing, fault tolerance, and horizontal scalability.

Simple architecture:

Data Sources → HDFS → YARN → MapReduce/Processing → Results → Analytics

The key business advantage is that the organization can process very large volumes of data by adding more machines instead of continuously upgrading one powerful machine.

